FINM3013 Introduction to Financial Derivatives

Chapter 11 Exercises

1. Draw a diagram showing the variation of an investor's profit and loss with the terminal stock price for a portfolio consisting of
    a. One share and a short position in one call option
    b. Two shares and a short position in one call option
    c. One share and a short position in two call options
    d. One share and a short position in four call options

    In each case, assume that the call option has an exercise price equal to the current stock price.

2. A call with a strike price of $60 costs $6. A put with the same strike price and expiration date costs $4. Construct a table that shows the profit from a straddle. For what range of stock prices would the straddle lead to a loss?

3. What trading position is created from a long strangle and a short straddle when both have the same time to maturity? Assume that the strike price in the straddle is halfway between the two strike prices of the strangle.